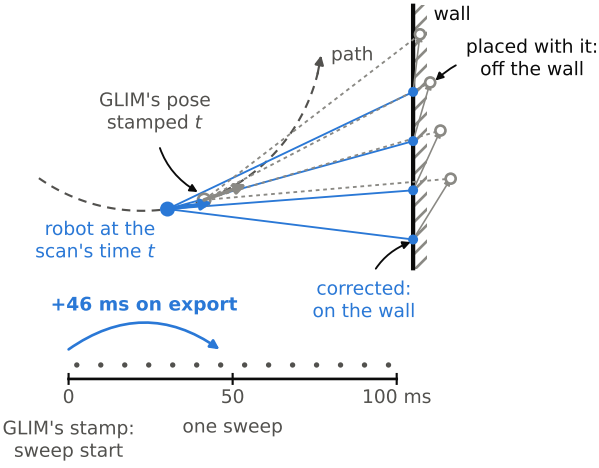
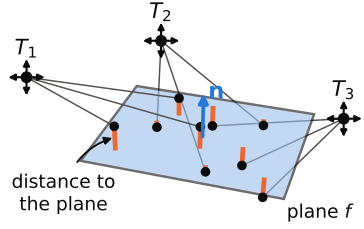
## (a) Pose-time correction




## (b) Plane bundle adjustment



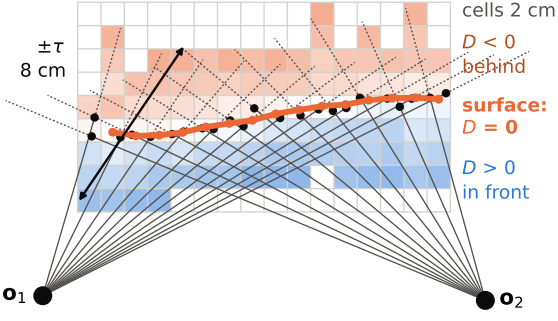

keyframe: 1 s of scans
plane: a flat 0.5 m cube
seen by ≥ 3 keyframes

plane in closed form (**n**, $d$);
all poses $T_k$ moved together:

$$\min_{T} \sum_{f} \lambda_{\min}(\mathrm{Cov}_f)$$

every plane's thickness$^2$

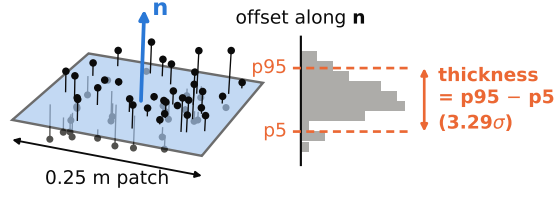
## (c) TSDF fusion




each cell within τ of a hit: $d = r - \|\mathbf{c}_v - \mathbf{o}\|$
$D(v)$ = mean of $d$ over all rays (at least 3)

## (d) Wall thickness

fit a plane to the patch's points




points as placed from one scan, 1 s or the whole run;
each scan against its own plane: the sensor alone